

Assignment 2 Solutions

$$2.2 \quad 4 \begin{bmatrix} -1 \\ 1 \\ 10 \end{bmatrix} - 5 \begin{bmatrix} -6 \\ 3 \\ 7 \end{bmatrix} + 9 \begin{bmatrix} 8 \\ 9 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 + 30 + 72 \\ 4 - 15 + 81 \\ 40 - 35 + 9 \end{bmatrix} = \begin{bmatrix} 98 \\ 70 \\ 14 \end{bmatrix}$$

$$2.4 \quad x_1 \begin{bmatrix} 4 \\ 10 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 4 \end{bmatrix} + x_3 \begin{bmatrix} 9 \\ -6 \end{bmatrix} + x_4 \begin{bmatrix} -10 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -8 \end{bmatrix}$$

$$2.7 \quad \text{Not possible, } \vec{v}_1 = \begin{bmatrix} -12 \\ -8 \end{bmatrix} = -4 \begin{bmatrix} 3 \\ 2 \end{bmatrix} = -4 \vec{v}_2$$

$$\text{so } \text{span} \{v_1\} = \text{span} \{v_2\}$$

$$2.8 \quad \vec{v}_1 + \vec{v}_2 = \begin{bmatrix} -7 \\ -1 \\ 0 \end{bmatrix} + \begin{bmatrix} -8 \\ 9 \\ -7 \end{bmatrix} = \begin{bmatrix} -7 + -8 \\ -1 + 9 \\ 0 + -7 \end{bmatrix} = \begin{bmatrix} -15 \\ 8 \\ -7 \end{bmatrix}$$

This vector is in $\text{span} \{\vec{v}_1, \vec{v}_2\}$ by def.

It's not in $\text{span} \{\vec{v}_1\}$ or $\text{span} \{\vec{v}_2\}$ since it's not a scalar multiple of $\vec{v}_1$ or $\vec{v}_2$

$$2.9 \quad \begin{bmatrix} -2 & -6 & 2 & 0 \\ -3 & -9 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} -6 & -18 & 6 & 0 \\ 6 & 18 & -2 & -8 \end{bmatrix}$$

$$\sim \begin{bmatrix} -6 & -18 & 6 & 0 \\ 0 & 0 & 4 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & -1 & 0 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 0 & -2 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

$$x_1 = -2 - 3x_2$$

x_2 is free

$$x_3 = -2$$

2.10

$$\begin{bmatrix} -1 & -1 & -2 & -2 \\ -2 & -3 & -9 & -10 \\ 1 & -2 & -13 & -16 \end{bmatrix} \sim \begin{bmatrix} -1 & -1 & -2 & -2 \\ 0 & -1 & -5 & -6 \\ 0 & -3 & -15 & -18 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & -1 & -2 & -2 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} -1 & 0 & 3 & 4 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 0 & -3 & -4 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = -4 + 3x_3$$

$$x_2 = 6 - 5x_3$$

x_3 is free

2.12

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -3 \\ 3 & 8 & -20 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -3 \\ 0 & 5 & -20 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & -5 \end{bmatrix}$$

not in span, corresponding vector equation has no solution

2.17

$$\begin{bmatrix} 1 & 1 & b_1 \\ 0 & 1 & b_2 \\ -2 & 1 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & b_1 \\ 0 & 1 & b_2 \\ 0 & 3 & b_3 + 2b_1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & b_1 \\ 0 & 1 & b_2 \\ 0 & 0 & b_3 + 2b_1 - 3b_2 \end{bmatrix}$$

$$2x_1 - 3x_2 + x_3 = 0$$

2.20

True, $\vec{u} = -\vec{v}_1 - \vec{v}_2$

2.21

False, $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

2.22 False, $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$

2.23 False, $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\vec{v}_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ $\vec{v}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

2.25

$$\begin{bmatrix} 1 & 0 & b_1 \\ 2 & 1 & b_2 \\ -1 & 1 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 - 2b_1 \\ 0 & 1 & b_3 + b_1 \end{bmatrix} \xrightarrow{-(b_2 - 2b_1)}$$

$$\sim \begin{bmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 - 2b_1 \\ 0 & 0 & b_3 + b_1 - b_2 + 2b_1 \end{bmatrix} \quad 3x_1 - x_2 + x_3 = 0$$

$$\begin{bmatrix} 1 & 0 & b_1 \\ 1 & 1 & b_2 \\ 1 & 3 & b_3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 - b_1 \\ 0 & 3 & b_3 - b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & b_1 \\ 0 & 1 & b_2 - b_1 \\ 0 & 0 & b_3 - 3b_2 + 2b_1 \end{bmatrix}$$

$$2x_1 - 3x_2 + x_3 = 0$$

$$3x_1 - x_2 + x_3 = 0$$

$$2x_3 - 3x_2 + x_3 = 0$$

$$\begin{bmatrix} 3 & -1 & 1 \\ 2 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} 6 & -2 & 2 \\ -6 & 9 & -3 \end{bmatrix} \sim \begin{bmatrix} 3 & -1 & 1 \\ 0 & 7 & -1 \end{bmatrix} \sim \begin{bmatrix} 3 & -1 & 1 \\ 0 & 1 & -1/7 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & 0 & 6/7 \\ 0 & 1 & -1/7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2/7 \\ 0 & 1 & -1/7 \end{bmatrix}$$

$$x_1 = -\frac{2}{7} x_3$$

$$x_2 = \frac{1}{7} x_3$$

x_3 is free

$$\begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$$